

Apiwich Sumeksri

(416) 930 3290 | apiwich.sumeksri@mail.utoronto.ca | [linkedin.com/in/apiwichsumeksri](https://www.linkedin.com/in/apiwichsumeksri) | github.com/apiwichs

Personal website: apiwichs.github.io

Previous **Hardware Engineering Intern at Analog Devices**. Third-year ECE student at the University of Toronto, seeking a 12–16 month PEY placement in analog, mixed-signal, embedded, or FPGA hardware.

EDUCATION

University of Toronto

Sept. 2024 – Jun. 2028

BASc, Electrical and Computer Engineering

- 3x Dean's Honour List (Fall 2024, Winter 2025, Winter 2026)
- **Relevant coursework:** Electronic Circuits (A), Circuit Analysis (A), Computer Organization (A); Digital Systems, Signals & Systems

TECHNICAL SKILLS

Languages: C, C++, Python (NumPy, Pandas, Matplotlib), Verilog, RISC-V Assembly, Java

Software & Tools: Altium Designer, LTspice, PyLTSpice, ModelSim, Quartus, STM32CubeIDE, Digilent Waveforms, Git

Protocols: SPI, I2C, UART

EXPERIENCE

Analog Devices, Inc.

May 2026 – Sept. 2026

Hardware Engineering Intern

- Extended the output range of a precision source-measure unit (SMU) card from **$\pm 50\text{V}$ to $\pm 100\text{V}$ at 200mA** , owning the high-voltage output stage end to end. The card closes two feedback paths: voltage feedback regulating at the DUT pin, current feedback clamping at the IDAC-programmed limit.
- Built and brought up a protoboard of the extended drive stage with both feedback paths closed, demonstrating **$\pm 98\text{V}$ measured output** on $\pm 100\text{V}$ rails and correlating bench results against simulation across the full range.
- Designed a **cascode output stage** so no single MOSFET sees the full $200\text{V } V_{\text{DS}}$ at 200mA , and selected the supporting high-voltage devices: complementary MOSFET pairs, 300V -class BJTs, and precision op-amps.
- Recalculated the DC biasing network for the doubled rail, re-deriving operating points across the push-pull output stage, cascode devices, and Zener/BJT bias spreader to keep every device in its linear region.
- Verified designs in LTspice, automating regression testing in Python (**PyLTSpice**) to sweep DC input across schematic revisions and flag divergence by percent-difference tolerance. Traced a DC convergence failure blocking simulation of the stage to floating nodes in the negative pull-down sub-circuit.
- Characterized the original $\pm 50\text{V}$ card on **Credence TMT ASL1000** production ATE to establish the performance baseline for the extended design.

University of Toronto Thai Student Association

Jun. 2025 – Present

Events Associate

- Built and queried an SQL-based attendee database to group registrants by event, arrival time, and role, enabling faster check-in and smoother on-site coordination.
- Acted as point of contact between executives, volunteers, and attendees during events, resolving real-time issues and adjusting task assignments to maintain smooth operations.

Provectus Networks

Nov. 2022 – Mar. 2023

IT Engineering Intern

- Upgraded client and institutional workstations by replacing outdated DDR3 RAM and hardware components, **reducing boot and loading times by up to 35%**.
- Diagnosed and resolved technical issues including hardware failures, driver conflicts, and OS errors across 20+ devices.

Hexforge Arena

Digital Systems Designer

- Built a real-time top-down PvP fighting game in bare-metal C on the DE1-SoC with no OS or graphics libraries, directly interfacing with memory-mapped VGA for all rendering.
- Implemented a dynamic camera system tracking both players and adjusting zoom based on distance, using nearest-neighbor sampling for manual pixel scaling to the VGA buffer.
- Designed class-based combat (Warrior and Sorceress) with animation state machines, hit detection, and real-time input handling, all coordinated within a single bare-metal loop.

CPU Pipeline Performance Simulator

Systems Programmer

- Developed a cycle-accurate 5-stage CPU pipeline simulator modeling IF/ID/EX/MEM/WB execution, instruction retirement, and pipeline drain behavior.
- Implemented RAW data hazard detection and stall insertion logic to simulate pipeline bubbles and dependency-driven performance penalties.
- Analyzed pipeline performance by computing CPI, stall rate, and stage occupancy from cycle-level execution data, visualizing results with Python (NumPy, Pandas, Matplotlib).

Wayout 1984-Inspired FPGA Maze Game

Digital Systems Designer

- Delivered an FPGA-based first-person maze game over a 3-week development cycle, integrating clock dividers, pixel-timing generators, and VGA output.
- Developed control logic using shift registers, finite state machines (FSMs), and synchronous update cycles to manage player movement, wall detection, and state transitions.
- Verified system behavior through ModelSim waveform analysis, register-level debugging, and cycle-accurate simulation to ensure reliable frame timing, memory access, and collision-detection accuracy.

Soundproofing of Elevator-Adjacent Rooms

Project Manager

- Led a 5-member team on an acoustic noise-reduction design project, earning recognition as one of the top 28 projects out of 187 teams in the faculty showcase.
- Implemented structured Gantt-chart planning, improving workflow efficiency by 30% and enabling the team to deliver the final design 2 weeks ahead of schedule.
- Developed a requirements document translating stakeholder needs into functional specifications, performance targets, and engineering constraints, providing the foundation for all design decisions.